

# Bayesian meta-analysis with dependent effect sizes in R: A tutorial on the bayesRVE package

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**ABSTRACT.** Meta-analytic data in psychology typically contain several effect sizes per study, arising from multiple measures, informants, time points or treatment contrasts, and effect sizes from the same study are statistically dependent. Two approaches to this problem have developed largely separately. Robust variance estimation with small-sample corrections provides standard errors that remain valid when the assumed dependence structure is wrong, but offers no Bayesian machinery; Bayesian multilevel modeling provides shrinkage estimation and full posterior distributions, but its interval estimates are only as trustworthy as the assumed within-study covariance structure. We describe the `bayesRVE` package for R, which combines the two approaches: the point estimate is the Bayesian posterior mean, the interval width is governed by a bias-reduced (CR2) cluster-robust sandwich estimate, and the interval tails use per-coefficient Satterthwaite degrees of freedom. The workflow consists of four functions, `bm_fit()`, `sandwich()`, `ccc()` and `summary()`, and is illustrated in a complete analysis of a publicly available psychology dataset comprising 113 correlations between school-motivation risk and later delinquency from 17 samples, including meta-regression with moderators, Markov chain Monte Carlo diagnostics, and the package’s small-sample guardrails. We show that the calibrated interval does not depend on an assumed within-study correlation, describe cross-checks against the `metafor` and `clubSandwich` packages, and suggest what to report. All data and code are publicly available, and the analyses reproduce in about a minute on a standard laptop.

**Keywords:** meta-analysis; dependent effect sizes; robust variance estimation; Bayesian inference; cluster-robust standard errors; R

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## 1. Introduction

Meta-analyses in psychology commonly include several effect sizes from the same study. A trial may report an intervention effect on three outcome scales, a longitudinal study may correlate a risk factor with an outcome at several follow-up waves, and two treatment arms may be compared with a common control group. Effect sizes computed from the same sample are statistically dependent, and reviews of current practice indicate that such data structures are the rule rather than the exception in psychology and neighboring fields (López-López et al., 2018; Tipton et al., 2019). Standard random-effects meta-analysis assumes one independent effect size per study. When it is applied to dependent effect sizes, uncertainty is understated: confidence intervals are too narrow and type I error rates exceed their nominal levels (Hedges et al., 2010; Pustejovsky & Tipton, 2022; Van Den Noortgate et al., 2015).

Two approaches to this problem are in common use. The first is robust variance estimation (RVE), introduced to meta-analysis by Hedges et al. (2010). A plausible “working model” for the dependence is fitted, and cluster-robust (sandwich) standard errors are then computed that remain valid even when the working model is misspecified. With the small-sample corrections developed by Tipton (2015) and Tipton and Pustejovsky (2015), which combine bias-reduced CR2 adjustments (Bell & McCaffrey, 2002; Pustejovsky & Tipton, 2018) with Satterthwaite degrees of freedom (Satterthwaite, 1946), RVE provides reliable frequentist inference with as few as 10 to 20 studies. The approach is implemented in mature software: the **robumeta** package (Fisher et al., 2023), and the **metafor** package (Viechtbauer, 2010) in combination with **clubSandwich** (Pustejovsky, 2026), organized around the correlated-and-hierarchical-effects (CHE) working model of Pustejovsky and Tipton (2022). What this approach does not provide is Bayesian machinery. There is no shrinkage of noisy estimates toward the pooled mean and no principled way to incorporate prior information, and the heterogeneity parameters are estimated only incidentally, without accompanying uncertainty statements (Tanner-Smith et al., 2016).

The second approach is Bayesian multilevel meta-analysis, using general-purpose software such as **brms** (Bürkner, 2017) built on Stan (Carpenter et al., 2017), or more specialized frameworks such as robust Bayesian model-averaged meta-analysis (Bartoš et al., 2026). Bayesian estimation has well-known attractions for psychologists, including direct probability statements, stabilized small-sample estimates and full posterior distributions for heterogeneity parameters, and it is explicitly encouraged in the statistical guidelines of the Psychonomic Society. However, the posterior

standard deviation of the pooled effect is only as trustworthy as the likelihood that produced it. With dependent effect sizes the analyst must specify the within-study covariance structure, typically through an assumed sampling correlation  $\rho$  that primary studies almost never report (Jackson et al., 2011; Pustejovsky & Chen, 2024). If that structure is misspecified, the posterior interval is miscalibrated, sometimes too narrow and sometimes too wide, and nothing in the standard Bayesian output reveals the problem (Lee & Tipton, 2026a). Recent work on the frequentist properties of Bayesian meta-analysis documents this sensitivity (Mathur, 2025).

The `bayesRVE` package (Lee, 2026) combines elements of the two approaches. The point estimate is the Bayesian posterior mean, with its shrinkage intact. The interval width is governed by the CR2 cluster-robust sandwich variance. The interval tails use per-coefficient Satterthwaite  $t$  reference distributions. The device that assembles these components, the Cholesky calibration correction (CCC), is an affine transformation of the posterior draws that leaves their center untouched while adjusting their spread to match the robust target, so that the result is again a set of posterior draws that can be summarized, plotted and propagated in the usual way. The theory underlying this construction, its validation in simulation, and the associated diagnostics are developed in three companion papers (Lee & Tipton, 2026a, 2026b; Lee et al., 2026); none of the details are required to follow this tutorial. In the words of the package documentation, the calibrated interval is Bayesian in its center, sandwich-robust in its width, and small-sample corrected in its tails.

This paper is a tutorial, in the spirit of earlier RVE tutorials for applied audiences (Tanner-Smith & Tipton, 2014; Tanner-Smith et al., 2016), and is written for the analyst rather than the theoretician: what to install, what the functions do, how to read the output, and what to report. As a running example we reanalyze the teaching dataset from Tanner-Smith et al. (2016), comprising 113 correlations between school-motivation risk factors and later delinquent behavior nested in 17 independent samples, so that the calibrated-Bayesian workflow can be compared directly with the classic frequentist RVE tutorial on the same data.

`bayesRVE` is not intended to replace the existing software ecosystem. For frequentist RVE, `metafor` with `clubSandwich`, or `robumeta`, remain the standard tools, with cluster wild bootstrapping available for very small samples (Joshi et al., 2022). Where publication bias under model uncertainty is the primary concern, robust Bayesian model averaging (Bartoš et al., 2026) addresses a different question from the one considered here, and fully multivariate Bayesian modeling with known within-study covariances

is better served by **brms**. The setting in which **bayesRVE** is useful is the one where three conditions hold together: (a) the effect sizes are nested in studies that contribute unequal numbers of estimates; (b) a Bayesian point estimate is wanted, with shrinkage, priors and full posteriors for the heterogeneity parameters; and (c) interval coverage is required that does not depend on the working covariance model being correct, with reasonable behavior when the number of studies is small. The package interoperates with **metafor** and **clubSandwich** rather than competing with them, and we describe the bridges in both directions below.

The plan of the paper is as follows. Section 2 describes the working models and the three-layer calibration in outline, together with the conditions under which the method should and should not be used. Section 3 introduces the package and its four-verb workflow, and Section 4 presents a first analysis on a small simulated dataset. Section 5 contains the main worked example, including convergence diagnostics and a comparison of four analyses of the same data. Section 6 adds moderators and discusses the small-sample guardrails, Section 7 considers sensitivity to the assumed within-study correlation, and Section 8 describes interoperability with **metafor** and **clubSandwich**. Section 9 suggests what to report, and Section 10 closes with a discussion. Every number and console display in the paper was generated by a single documented script, listed in full in the Online Supplement that follows the references (Section S2); the supplement also covers installation, prior specification and sensitivity, the correlated-effects working model, the full  $\rho$  diagnostics, and a gallery of the package’s warning messages.

## 2. Working models and calibrated inference

### 2.1. Working models for dependent effect sizes

Let  $y_{ij}$  denote effect size  $i$  reported by study  $j$ , with known sampling variance  $v_{ij}$ , for  $j = 1, \dots, J$  studies contributing  $n_j$  effect sizes each,  $N = \sum_j n_j$  in total, and let  $K$  denote the number of fixed effects. Following the working-model tradition in RVE (Pustejovsky & Tipton, 2022), **bayesRVE** fits one of two Bayesian multilevel models. The correlated-and-hierarchical-effects (CHE) model, which we use throughout this paper, allows effect sizes to vary both between and within studies,

$$y_{ij} = \mathbf{x}_{ij}^\top \boldsymbol{\beta} + u_j + w_{ij} + e_{ij}, \quad u_j \sim N(0, \tau^2), \quad w_{ij} \sim N(0, \omega^2), \quad (1)$$

where  $e_{ij} \sim N(0, v_{ij})$  is the sampling error with known variance,  $\tau$  is the between-study standard deviation and  $\omega$  the within-study standard deviation. This is the

Bayesian analogue of the three-level model familiar from Konstantopoulos (2011) and Van Den Noortgate et al. (2015). The simpler correlated-effects (CE) model omits  $w_{ij}$  and estimates a single between-study component  $\tau^2$ , and is closer to the original RVE working model of Hedges et al. (2010). Guidance on choosing between the two is given in the Online Supplement (Section S4).

Default priors are weakly informative, with  $\beta \sim N(0, 1)$  elementwise and half- $t(3, 0, 0.5)$  distributions on  $\tau$  and  $\omega$  (Gelman, 2006). The priors in force are printed with every fit, and the supplement (Section S3) describes how to modify them and how to examine whether they influence the results.

## 2.2. Why the posterior interval requires repair

The model in equation 1 is a working model, and its most fragile ingredient is the dependence structure. Sampling errors from the same study are correlated, because the same participants produced them; the correlations are essentially never reported; and any single assumed structure is almost certainly wrong in detail. A Bayesian posterior computed under a misspecified likelihood can nevertheless have a useful center, since the posterior mean remains a stable shrinkage estimate. Its spread is another matter. The posterior standard deviation answers the question of how uncertain one would be if the working model were exactly true, which is not the question of interest when the working model is in doubt. This distinction between the quality of the center and the quality of the spread is the starting point of the calibrated-Bayes tradition (Little, 2012; Rubin, 1984), in which Bayesian procedures are evaluated by their frequentist properties and repaired where those properties fail.

Frequentist RVE repairs the spread by replacing the model-based variance of the point estimator with an empirical cluster-level sandwich estimate (Hedges et al., 2010), bias-corrected for small samples (CR2; Bell & McCaffrey, 2002; Pustejovsky & Tipton, 2018; Tipton, 2015) and paired with Satterthwaite degrees of freedom so that  $t$ -based intervals behave reasonably when  $J$  is small. `bayesRVE` imports this repair into the Bayesian output.

## 2.3. Three-layer calibration

Figure 1 summarizes the resulting interval. The calibration takes each of three components from the framework that estimates it best: the location is the Bayesian posterior mean  $\hat{\beta}_k$ , which is left unchanged; the spread is the CR2 cluster-robust standard error, which does not require the working dependence structure to be correct;

and the tails follow a per-coefficient Satterthwaite  $t$  reference distribution, so that intervals widen appropriately when the effective number of clusters is small.

The device that accomplishes this is the Cholesky calibration correction. Writing  $\boldsymbol{\psi}(s)$  for posterior draw  $s$  of the fixed effects,  $\hat{\boldsymbol{\psi}}$  for the posterior mean,  $\Sigma_{\text{MCMC}}$  for the posterior covariance matrix and  $V_{\text{CR2}}$  for the sandwich target, every draw is transformed as

$$\boldsymbol{\psi}^*(s) = \hat{\boldsymbol{\psi}} + L_2 L_1^{-1}(\boldsymbol{\psi}(s) - \hat{\boldsymbol{\psi}}), \quad L_1 = \text{chol}(\Sigma_{\text{MCMC}}), \quad L_2 = \text{chol}(V_{\text{CR2}}). \quad (2)$$

This affine map preserves the center of the draws exactly while making their covariance match  $V_{\text{CR2}}$  exactly (Lee & Tipton, 2026a). Because the output is again a set of draws, the usual posterior summaries, plots and derived quantities remain available; the draws simply carry robust spread. Intervals are then formed on the  $t$  scale with Satterthwaite degrees of freedom, and we refer to them as CCC- $t$  intervals. The justification for combining a Bayesian posterior with a frequentist sandwich target in this way is given in Lee et al. (2026), and simulation evidence in Lee and Tipton (2026a) indicates that CCC- $t$  intervals attain close-to-nominal coverage for average effects and for meta-regression slopes in the small- $J$ , dependent-data settings typical of psychology, where both the uncalibrated Bayesian interval and asymptotic RVE can fall short.

Two properties of the calibration are worth noting before we turn to the software. First, it is informative even when it changes little. For every coefficient the package reports the ratio  $r_k$  of the calibrated to the uncalibrated standard error, so the analyst learns how far off the working model’s uncertainty assessment was; the uncalibrated Bayesian analysis contains no such information. Second, the correction is selective. In a given dataset some coefficients may need substantial repair while others need none, a phenomenon analyzed in Lee and Tipton (2026b), and the  $r_k$  ratios show which is which.

## 2.4. Scope

The package enforces an explicit operating envelope rather than failing silently. Calibration of the fixed effects requires  $J \geq J_{\min} = \max(8, 2K + 2)$  studies; below this threshold a cluster-robust covariance estimated from so few clusters is itself too noisy to serve as a target, and the package returns the raw posterior for the fixed effects together with a message, rather than an unstable calibration. In addition, whenever a coefficient’s Satterthwaite degrees of freedom fall below 4, the package warns that inference for that coefficient may be unreliable, in line with standard RVE

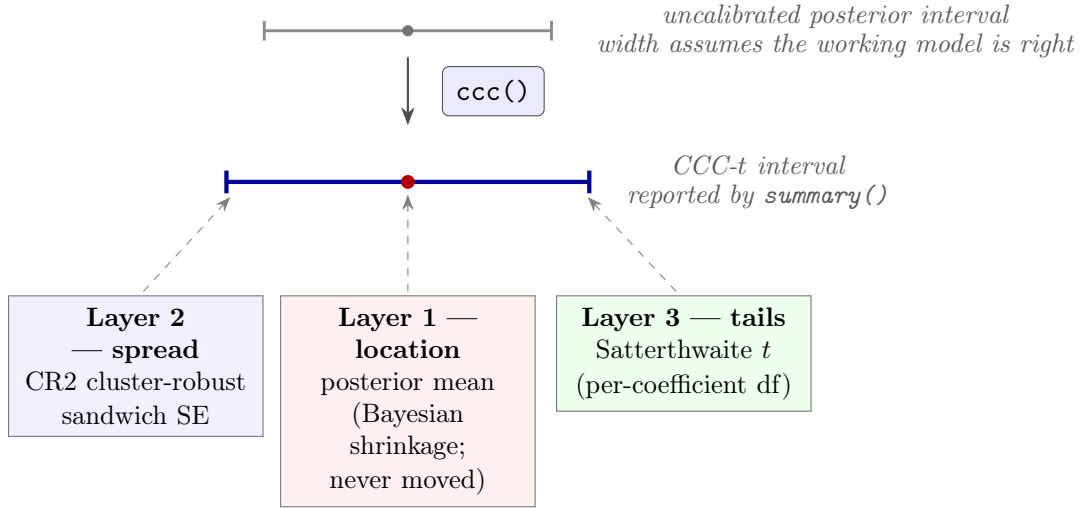


FIGURE 1. The three-layer calibration. The uncalibrated Bayesian interval (top, gray) has a trustworthy center but a model-dependent width. `ccc()` maps the posterior draws through equation 2: the center (layer 1) is preserved, the spread (layer 2) is replaced by the CR2 cluster-robust standard error, and the interval is read off a Satterthwaite- $t$  reference distribution (layer 3). Depending on the data, the calibrated interval can be narrower or wider than the uncalibrated one.

guidance (Tanner-Smith et al., 2016; Tipton, 2015). The appropriate responses are the same as in frequentist RVE: simplify the model, combine sparse categories, or treat the affected interval as descriptive. Both guardrails are illustrated below and in the supplement (Section S7).

The package is designed for the CE and CHE working models with a single level of clustering. It does not fit multivariate models with known within-study covariance matrices, selection models for publication bias (for which see Bartoš et al., 2026), or location-scale structures. We return to these boundaries in Section 10.

### 3. The bayesRVE package

#### 3.1. Installation

`bayesRVE` uses `CmdStan`, through the `cmdstanr` interface (Gabry et al., 2024), as its sampling backend. `CmdStan` is required only for fitting; all downstream steps are ordinary R. The one-time setup is

```
# 1. cmdstanr + CmdStan (one-time; only needed to FIT models)
install.packages("cmdstanr",
```

```

      repos = c("https://stan-dev.r-universe.dev",
                getOption("repos"))
cmdstanr::install_cmdstan()

# 2. bayesRVE from GitHub
# install.packages("pak")
pak::pak("joonho112/bayesRVE")

```

The first model fitted in a fresh installation compiles the Stan program once and caches the binary; subsequent fits on datasets of the size considered here take a few seconds. The supplement (Section S1) covers the C++ toolchain on each operating system and the cache-management helpers `bm_compile()` and `bm_clear_cache()`. This tutorial additionally uses three packages available from CRAN:

```
install.packages(c("metadat", "metafor", "clubSandwich"))
```

### 3.2. Workflow

Figure 2 shows the package as a whole. The pipeline is

```
bm_fit() → sandwich() → ccc() → summary()
```

and only the first step involves Stan. `bm_fit()` samples the working model; `sandwich()` constructs the CR2 robust target and the Satterthwaite degrees of freedom from the fitted object; `ccc()` applies the calibration of equation 2; and the usual accessors read out the results. Because the three downstream steps are inexpensive computations on the stored draws, they can be rerun freely, so that alternative sandwich options, recalibrations and sensitivity analyses all reuse a single MCMC run. Table 1 lists the exported functions by workflow stage.

### 3.3. Data layout and model specification

The package expects data in the long format that RVE users already keep, with one row per effect size, a column for the effect estimate, a column for its sampling variance, and a column identifying the study (cluster). The model is declared with an extended formula whose left-hand side names the effect and its variance:

```

bm_fit(yi | se(vi) ~ 1,      # intercept-only: average effect
       data = mydata,
       study = ~ studyid,    # the clustering variable
       model = "che")        # or "ce"

```

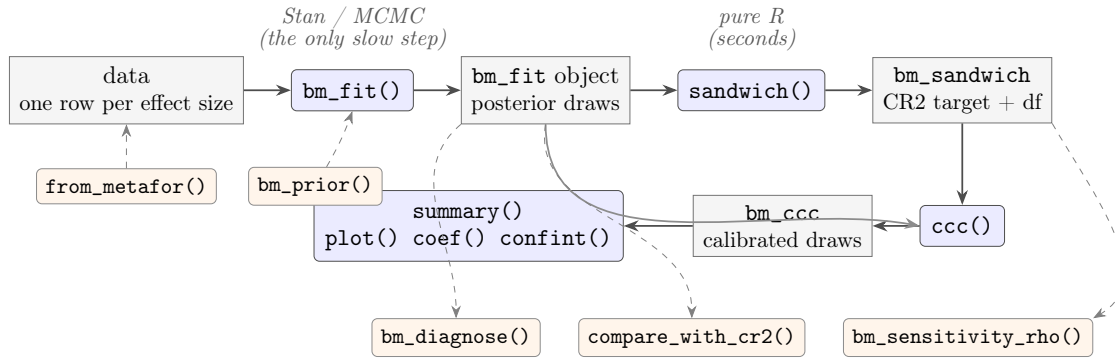


FIGURE 2. The `bayesRVE` workflow. Along the main pipeline (solid arrows), `bm_fit()` samples the Bayesian working model (the only step that requires CmdStan), `sandwich()` constructs the CR2 cluster-robust target with Satterthwaite df, `ccc()` recalibrates the posterior draws, and standard accessors read out the results. Helper functions (dashed) attach where they act: priors and `metafor` imports feed the fit, and diagnostics, cross-checks and  $\rho$  diagnostics read from fitted objects.

TABLE 1. The exported functions of `bayesRVE` (v0.7.0), by workflow stage.

| Stage              | Function(s)                                                                                                                                                                                                                                           | Purpose                                                                                                                                    |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Fit                | <code>bm_fit(formula, data, study, model)</code><br><code>bm_prior()</code> , <code>normal()</code> , <code>half_t()</code> ,<br><code>half_normal()</code> , <code>half_cauchy()</code><br><code>bm_compile()</code> , <code>bm_clear_cache()</code> | Sample the CE or CHE working model<br>Specify priors (optional; defaults are printed<br>with every fit)<br>Manage the compiled-model cache |
| Check              | <code>bm_diagnose()</code> , <code>bm_diagnose_all()</code>                                                                                                                                                                                           | MCMC diagnostics: divergences, $\hat{R}$ , ESS                                                                                             |
| Robust target      | <code>sandwich(fit, rho, small_sample)</code>                                                                                                                                                                                                         | CR2 (default), scalar, or raw sandwich + df                                                                                                |
| Calibrate          | <code>ccc(fit, sandwich_obj)</code>                                                                                                                                                                                                                   | Three-layer calibration of the draws                                                                                                       |
| Read out           | <code>summary()</code> , <code>plot()</code> , <code>coef()</code> ,<br><code>confint()</code> , <code>vcov()</code> , <code>nobs()</code>                                                                                                            | Accessors on fitted and<br>calibrated objects                                                                                              |
| $\rho$ diagnostics | <code>bm_sensitivity_rho()</code> , <code>_rho_study()</code> ,<br><code>_rho_table()</code> , <code>bm_find_rho_implied()</code>                                                                                                                     | Design-effect-ratio decomposition; the<br>correlation implied by the data                                                                  |
| Interop            | <code>from_metafor()</code> , <code>compare_with_cr2()</code>                                                                                                                                                                                         | Import an <code>rma.mv</code> workflow; cross-check<br>against <code>clubSandwich</code>                                                   |

Moderators enter on the right-hand side as in `lm()`, for example `yi | se(vi) ~ age + sexmix`. Despite its name, `se()` takes the sampling variance column, mirroring the `vi` convention of `metafor`'s `escalc()` datasets; the datasets used in this tutorial all ship with `vi` precomputed.

## 4. A first analysis

The package ships a small simulated dataset, `example_meta`, containing 93 standardized-mean-difference effect sizes from 20 hypothetical studies of an education intervention, together with a treatment-type moderator. Its purpose is to let users verify their installation and become familiar with the output format before turning to real data.

```
library(bayesRVE)
data(example_meta)
head(example_meta)
```

|   | studyid  | esid | yi     | vi     | tx_type  | study_year |
|---|----------|------|--------|--------|----------|------------|
| 1 | study_01 | 1    | 0.2328 | 0.1030 | direct   | -6         |
| 2 | study_01 | 2    | 0.0200 | 0.1166 | direct   | -6         |
| 3 | study_01 | 3    | 0.1689 | 0.0301 | direct   | -6         |
| 4 | study_02 | 1    | 0.7842 | 0.1602 | indirect | -4         |
| 5 | study_02 | 2    | 0.7168 | 0.0354 | indirect | -4         |
| 6 | study_02 | 3    | 0.6315 | 0.0838 | indirect | -4         |

We fit the CHE working model for the average effect. The first call in a fresh session prints Stan's sampling progress, which we elide here:

```
fit <- bm_fit(yi | se(vi) ~ 1,
              data = example_meta,
              study = ~ studyid,
              model = "che",
              seed = 42)
```

```
i Preparing data...
i Using Stan model "bm_che_ncp".
i Loading cached "bm_che_ncp" from disk.
i Sampling: 4 chains, 1000 warmup + 1000 sampling iterations each.
Running MCMC with 4 parallel chains...

## [... chain-by-chain progress omitted ...]

All 4 chains finished successfully.
Mean chain execution time: 0.5 seconds.
Total execution time: 0.7 seconds.

✓ Model "bm_che_ncp" fitted in 2.3s (20 studies, 1 fixed effect).
```

The downstream pipeline is three lines, none of which reruns Stan:

```
sw <- sandwich(fit)           # CR2 robust target (layer 2)
cal <- ccc(fit, sandwich_obj = sw) # recalibrate the draws
summary(cal)                  # layers 1-3 in one table
```

```
-- bayesRVE CCC Summary -----
K = 1 fixed effect, J = 20 studies, S = 4000 posterior draws
Sandwich: CR2 | df: satterthwaite | Inference: CCC-t (Three-Layer Calibration)
Center: posterior_mean | Level: 0.95
-- Beta Block (Cholesky CCC) -----
Frobenius ||A - I||_F: 0.03
Parameter Estimate SE    df  CI_95%_lo CI_95%_hi r_k  Direction
beta[1]   0.316    0.067 18.7 0.176    0.456    0.970 narrower
Beta r_k: 0 wider | 1 narrower | 0 ~same
-- Theta Block -----
Study effects analytically integrated out (CHE model).
No theta-block calibration applied.
-- Verification -----
✓ 5/6 checks passed (1 skipped: beta fallback).
```

The row in the beta block is the result that would be reported. The average effect is 0.316, the posterior mean (layer 1); its standard error of 0.067 is the CR2 sandwich estimate (layer 2); and the 95% interval [0.176, 0.456] is based on a  $t$  distribution with 18.7 Satterthwaite degrees of freedom (layer 3). Two further columns are specific to this framework. The ratio  $r_k = 0.970$  is the calibrated standard error divided by the uncalibrated posterior standard deviation; a value near 1, as here, indicates that the working model’s uncertainty assessment was already nearly correct, and the Frobenius norm near zero indicates that the calibration map as a whole moved the draws very little. In CHE summaries the coefficients are labeled `beta[1]`, `beta[2]`, ... in the order in which they enter the formula, intercept first. The verification block reports internal algebraic checks of the calibration, namely that the center is preserved and the covariance matched. The skipped check compares the full covariance matrix and applies only when  $K \geq 2$ , so with a single coefficient five checks run and one is skipped; the label “beta fallback” refers to the internal code path involved, not to any problem with the fit.

An installation that reproduces this table, with any differences confined to the third decimal place by MCMC noise across platforms, is ready for real data.

## 5. Application: school motivation and delinquency

### 5.1. Data

The running example is the teaching dataset assembled by Tanner-Smith et al. (2016) for their frequentist RVE tutorial and distributed as `dat.tannersmith2016` in the `metadat` package (Viechtbauer et al., 2026). The effect sizes are Fisher  $z$ -transformed correlations between school-motivation risk factors, such as low academic aspirations or low academic self-efficacy, measured between ages 11 and 18, and chargeable delinquent or criminal behavior measured at least six months later. Because the primary studies operationalized both constructs in several ways, each sample contributes several correlations, which is the correlated-effects structure of interest here. As in the original tutorial, the data are used for illustration only; they are a subset of a larger database and do not constitute a current synthesis of this literature.

```
library(bayesRVE)
library(metadat)

data(dat.tannersmith2016)
dat <- dat.tannersmith2016
head(dat, 8)
```

|   | studyid | yi     | vi    | sei   | aget1 | aget2 | propmale | sexmix |
|---|---------|--------|-------|-------|-------|-------|----------|--------|
| 1 | 2       | -0.060 | 0.001 | 0.027 | 14.5  | 18.0  | 1        | male   |
| 2 | 2       | -0.050 | 0.001 | 0.027 | 14.5  | 18.0  | 1        | male   |
| 3 | 3       | -0.050 | 0.001 | 0.028 | 14.5  | 18.0  | 0        | female |
| 4 | 3       | -0.050 | 0.001 | 0.028 | 14.5  | 18.0  | 0        | female |
| 5 | 5       | -0.010 | 0.006 | 0.080 | 11.5  | 13.5  | 1        | male   |
| 6 | 5       | 0.030  | 0.006 | 0.080 | 11.5  | 13.5  | 1        | male   |
| 7 | 5       | 0.100  | 0.006 | 0.080 | 11.5  | 13.5  | 1        | male   |
| 8 | 6       | 0.141  | 0.007 | 0.085 | 11.5  | 13.5  | 0        | female |

The dataset comprises  $N = 113$  effect sizes nested in  $J = 17$  independent samples (hereafter “studies”), with between 2 and 12 effect sizes per study and a median of 7. Figure 3 plots all 113 estimates by study. The clustering is evident: some studies contribute a tight group of similar correlations while others contribute a dozen more dispersed ones, and treating the 113 rows as independent would clearly overstate the information in the data. With 17 studies, small-sample behavior rather than

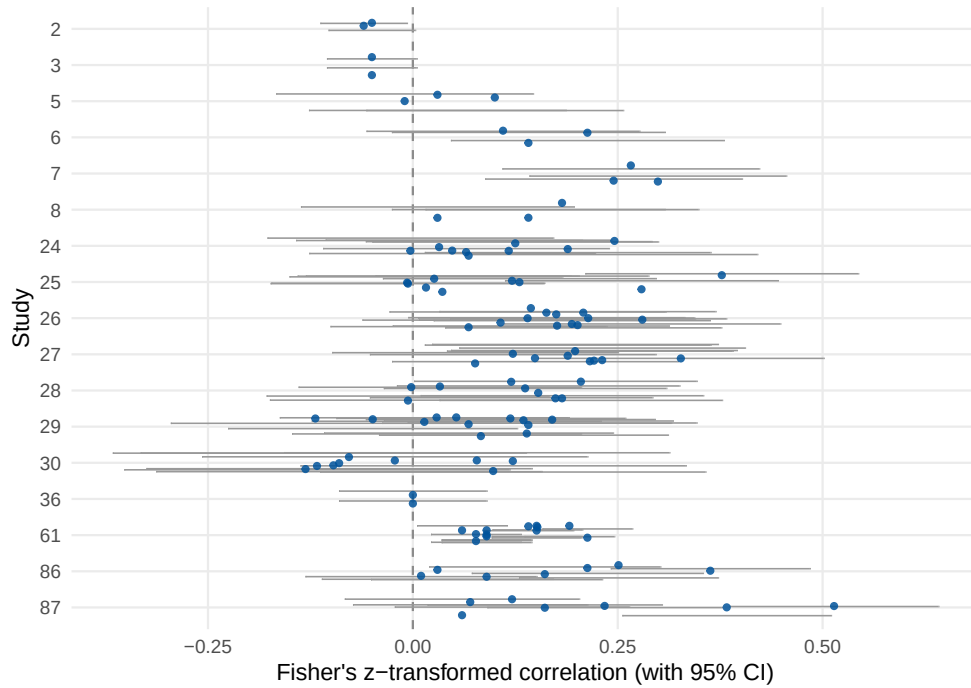


FIGURE 3. The running example: 113 Fisher- $z$  correlations between school-motivation risk and later delinquency, grouped by contributing study (Tanner-Smith et al., 2016; Viechtbauer et al., 2026). Horizontal lines are per-estimate 95% CIs; vertical scatter within a study row separates overlapping estimates.

asymptotics governs the inference, which is the setting the three-layer calibration is intended for.

## 5.2. Fitting the working model

```
fit <- bm_fit(yi | se(vi) ~ 1,
             data = dat,
             study = ~ studyid,
             model = "che",
             seed = 42)
print(fit)
```

```
-- bayesRVE Model Fit
Call: bm_fit(formula = yi | se(vi) ~ 1, data = dat, study = ~studyid, model =
"che", seed = 42)
Model: "bm_che_ncp" (Non-Centered (NCP))
Studies (J): 17 | Observations (N): 113 | Fixed effects (K): 1
```

```

Prior: beta ~ Normal(mean = 0, sd = 1), tau ~ Half-t(df = 3, location = 0, scale =
0.5)
MCMC: 4 chains, 4000 total draws
Elapsed: 2s

```

The print method restates what was fitted, including the model, the sample sizes, the priors and the sampler settings, and provides the record needed for a Methods section. The posterior summary shows what the Bayesian model alone concludes:

```

summary(fit)

-- bayesRVE Model Summary -----
Call: bm_fit(formula = yi | se(vi) ~ 1, data = dat, study = ~studyid, model =
"che", seed = 42)
Studies (J): 17 | Observations (N): 113 | Fixed effects (K): 1
Parameterization: ncp | Inference: Posterior (uncalibrated)
Center: posterior_mean | Level: 0.95
-- Fixed Effects (beta) -----
Parameter   Estimate SE    CI_95%_lo CI_95%_hi Rhat  ESS
(Intercept) 0.102   0.025 0.053     0.151     1.002 2484
-- Heterogeneity (tau) -----
Mean: 0.09 | Median: 0.087 | SD: 0.022
95% CI: [0.053, 0.141]
-- Within-study heterogeneity (omega) -----
Mean: 0.05 | Median: 0.051 | SD: 0.011
95% CI: [0.028, 0.073]
-----

```

The header identifies this display as the uncalibrated posterior. Two elements are best read from this display rather than from the calibrated one. The first is the pair of heterogeneity blocks: the between-study standard deviation  $\tau$  has posterior mean 0.09 and the within-study standard deviation  $\omega$  has posterior mean 0.05, each with a credible interval. In frequentist RVE these variance components are incidental quantities estimated without uncertainty statements (Tanner-Smith et al., 2016), whereas here they are ordinary parameters with full posteriors. The clearly nonzero  $\omega$  also supports the choice of the CHE over the CE working model for these data. The second element is the pair of MCMC columns ( $\hat{R}$ , ESS), which give a first indication that sampling went well; the dedicated diagnostic makes this check explicit.

### 5.3. Convergence diagnostics

Results from a sampler that has not converged should not be reported, whether calibrated or not. `bm_diagnose()` applies the standard checks (Vehtari et al., 2021) and states its conclusion:

```
bm_diagnose(fit)

-- bayesRVE MCMC Diagnostics -----
✓ Status: OK
Chains: 4 | Warmup: 1000 | Sampling: 1000
Backend: posterior package
✓ Divergent transitions: 0
✓ Max treedepth hits: 0
✓ E-BFMI per chain: 0.832, 1.015, 1.02, 0.946

Rhat summary (all 19 parameters):
min = 1.001 | median = 1.001 | max = 1.001
✓ All Rhat <= 1.01

ESS bulk summary:
min = 2448 | Q1 = 2526 | median = 2604
✓ All ESS_bulk >= 400

ESS tail summary:
min = 1996 | Q1 = 2032 | median = 2069
✓ All ESS_tail >= 400
-----
```

With no divergent transitions, all  $\hat{R} \leq 1.01$  and effective sample sizes in the thousands, we proceed. If this check fails, the calibration should not be run; the supplement (Section S1) lists the remedies in the order in which we suggest trying them (more iterations, a higher `adapt_delta`, and a review of model and priors), all of which are arguments to `bm_fit()`. One benign message is worth mentioning because it often alarms new Stan users: occasional notes during warmup that a “Metropolis proposal is about to be rejected” are a normal part of early adaptation, as the message text itself indicates, and require no action when they occur sporadically.

### 5.4. Calibrated inference

```
sw <- sandwich(fit, rho = 0.8) # CR2 target (rho: see Section 7)
```

```
cal <- ccc(fit, sandwich_obj = sw) # three-layer calibration
summary(cal)
```

```
-- bayesRVE CCC Summary -----
K = 1 fixed effect, J = 17 studies, S = 4000 posterior draws
Sandwich: CR2 | df: satterthwaite | Inference: CCC-t (Three-Layer Calibration)
Center: posterior_mean | Level: 0.95
-- Beta Block (Cholesky CCC) -----
Frobenius ||A - I||_F: 0.102
Parameter Estimate SE    df  CI_95%_lo CI_95%_hi r_k  Direction
beta[1]   0.102    0.022 15.9 0.054     0.149    0.898 narrower
Beta r_k: 0 wider | 1 narrower | 0 ~same
-- Theta Block -----
Study effects analytically integrated out (CHE model).
No theta-block calibration applied.
-- Verification -----
✓ 5/6 checks passed (1 skipped: beta fallback).
```

The average  $z$ -transformed correlation is estimated as 0.102, with CR2 standard error 0.022 and 95% CCC- $t$  interval [0.054, 0.149] on 15.9 Satterthwaite degrees of freedom. Back-transformed, this corresponds to a correlation of about .10 with 95% confidence interval [.05, .15]: school-motivation risk shows a small positive association with later delinquency, in agreement with the substantive conclusion of Tanner-Smith et al. (2016) for these data.

The diagnostic columns add information that the frequentist analysis does not provide. Here  $r_k = 0.898$ , so the uncalibrated posterior standard deviation (0.025) was about 11% larger than the robust standard error (0.022); for this coefficient the working model somewhat overstated the uncertainty, and the calibration tightened the interval. The direction and magnitude of the correction are empirical results that differ across coefficients and datasets, and we caution against expecting intervals always to narrow; a case in which the uncalibrated interval is too narrow by a factor of two appears in Section 6.2.

For programmatic access, the accessors return the layers separately: `coef()` gives the posterior means (layer 1), `vcov()` the calibrated covariance (layer 2), and `confint()` the assembled intervals, whose `df` attribute records layer 3.

```
coef(cal)
confint(cal)
```

TABLE 2. Four analyses of the average school-motivation–delinquency correlation (Fisher- $z$  scale). All rows analyze the same  $N = 113$  effect sizes from  $J = 17$  studies. The first row is included as a cautionary baseline only.

| Analysis                                        | Est.  | SE    | df   | 95% CI         | Width |
|-------------------------------------------------|-------|-------|------|----------------|-------|
| Ignore dependence (RE on all rows) <sup>a</sup> | 0.116 | 0.011 | —    | [0.094, 0.137] | 0.043 |
| CHE + CR2 RVE (metafor/clubSandwich)            | 0.102 | 0.024 | 14.9 | [0.051, 0.154] | 0.103 |
| Bayes CHE, uncalibrated posterior               | 0.102 | 0.025 | —    | [0.053, 0.151] | 0.098 |
| Bayes CHE + CCC- $t$ (bayesRVE)                 | 0.102 | 0.022 | 15.9 | [0.054, 0.149] | 0.094 |

*Note.* Est. = point estimate; CI = confidence/credible interval; Width = upper minus lower CI bound; df = Satterthwaite degrees of freedom where applicable. Reproduced by the replication script (Supplement S2).

<sup>a</sup> REML random-effects model `rma(yi, vi)` treating the 113 effect sizes as independent.

```

    beta[1]
0.1015828

          2.5 %    97.5 %
beta[1] 0.05440138 0.1487642
attr(,"df")
[1] 15.88111
attr(,"method")
[1] "CCC-t (Satterthwaite df, per-coefficient)"

```

### 5.5. Comparison with alternative analyses

Table 2 collects four analyses of the same 113 correlations. The first row is a standard random-effects model that treats all rows as independent; it is included only as a cautionary baseline, since this analysis is never appropriate for such data. Its interval, [0.094, 0.137], is less than half as wide as that of any analysis that respects the clustering, which is the familiar false precision of ignored dependence (Hedges et al., 2010; Van Den Noortgate et al., 2015) and the same contrast used by Tanner-Smith et al. (2016) to motivate RVE on these data. The second row is the current frequentist standard, the CHE working model in `metafor` with CR2 standard errors from `clubSandwich`. The third and fourth rows are the Bayesian model before and after calibration.

Three observations from Table 2 generalize beyond this example. First, every analysis that accounts for the dependence differs from the naive baseline by more than a factor of two in interval width, so that acknowledging the clustering at all

is by far the most consequential analytic decision. Second, the calibrated Bayesian analysis and the frequentist CHE + CR2 analysis agree closely here, with identical point estimates to three decimals and standard errors of 0.022 and 0.024; when two rather different repair strategies converge on the same answer, that agreement is itself a useful robustness check, and Section 8 shows how to produce it in a single call. Third, in this dataset the uncalibrated Bayesian interval was already close to correct (width 0.098, against 0.094 after calibration), but this could not have been known without carrying out the calibration; in the meta-regression of Section 6, the same working model applied to the same data understates one coefficient’s uncertainty by a factor of two. The calibration identifies which of these situations obtains.

The point estimates deserve brief comment. The Bayesian posterior mean (0.1016) and the REML estimate (0.1023) barely differ here, as is expected with  $J = 17$  studies and a reasonably strong signal. The advantages of the Bayesian center arise in less favorable settings, with fewer studies, noisier data or variance components near the boundary, where shrinkage stabilizes the estimate; simulation evidence on these settings is given in Lee and Tipton (2026a). In well-behaved problems, calibrated Bayes and small-sample RVE should agree, and here they do.

## 6. Meta-regression

### 6.1. A study-level moderator

We next ask whether the motivation–delinquency correlation varies with the age at which motivation was measured. Following Tanner-Smith et al. (2016), who aggregate participant age to the study level before using it as a moderator, we center the study-mean age and add it to the formula; nothing else in the workflow changes.

```
dat$aget1_study <- ave(dat$aget1, dat$studyid) # study-mean age
dat$age_c <- dat$aget1_study - mean(dat$aget1_study)

fit_mod <- bm_fit(yi | se(vi) ~ age_c,
                  data = dat,
                  study = ~ studyid,
                  model = "che",
                  seed = 42)
sw_mod <- sandwich(fit_mod, rho = 0.8)
cal_mod <- ccc(fit_mod, sandwich_obj = sw_mod)
summary(cal_mod)
```

```
| -- bayesRVE CCC Summary -----
```

```

K = 2 fixed effects, J = 17 studies, S = 4000 posterior draws
Sandwich: CR2 | df: satterthwaite | Inference: CCC-t (Three-Layer Calibration)
Center: posterior_mean | Level: 0.95
-- Beta Block (Cholesky CCC) -----
Frobenius ||A - I||_F: 0.385
Parameter Estimate SE    df  CI_95%_lo CI_95%_hi r_k  Direction
beta[1]   0.098    0.022 12.4  0.049    0.146    0.843 narrower
beta[2]  -0.013    0.010  6.1  -0.036    0.010    0.648 narrower
Beta r_k: 0 wider | 2 narrower | 0 ~same
-- Theta Block -----
Study effects analytically integrated out (CHE model).
No theta-block calibration applied.
-- Verification -----
✓ All 6 verification checks passed.
-----

```

Here `beta[1]` is the intercept, the average correlation at the mean age, and `beta[2]` is the age slope, estimated as  $-0.013$  per year with 95% interval  $[-0.036, 0.010]$  on 6.1 degrees of freedom. There is thus no reliable evidence of an age trend from this main-effect specification, in line with what Tanner-Smith et al. (2016) found for age main effects on these data. The `df` column behaves as the RVE literature leads one to expect: the intercept has 12.4 effective degrees of freedom but the slope only 6.1, because degrees of freedom in cluster-robust inference depend on the number of studies and on the balance and leverage of the covariate across studies, not on the 113 rows (Tanner-Smith et al., 2016; Tipton, 2015). The interval for the slope is accordingly based on a wider-tailed  $t$  distribution. Figure 4 shows the comparison plot for this model, produced by `plot(cal_mod)`: both coefficients keep their centers, and both intervals tighten moderately under calibration, with  $r_k$  of 0.843 and 0.648.

## 6.2. Small degrees of freedom

The moderator above was aggregated to the study level. Suppose that the raw, within-study-varying age variable had been used instead, a natural-looking choice that changes the leverage structure of the design considerably:

```

dat$age_raw_c <- dat$aget1 - mean(dat$aget1)
fit_raw <- bm_fit(yi | se(vi) ~ age_raw_c, data = dat,
                 study = ~ studyid, model = "che", seed = 42)
sw_raw <- sandwich(fit_raw, rho = 0.8)

```

```

Warning: ! Satterthwaite df = 3.95 < 4 for coefficient 2; inference may be unreliable

```

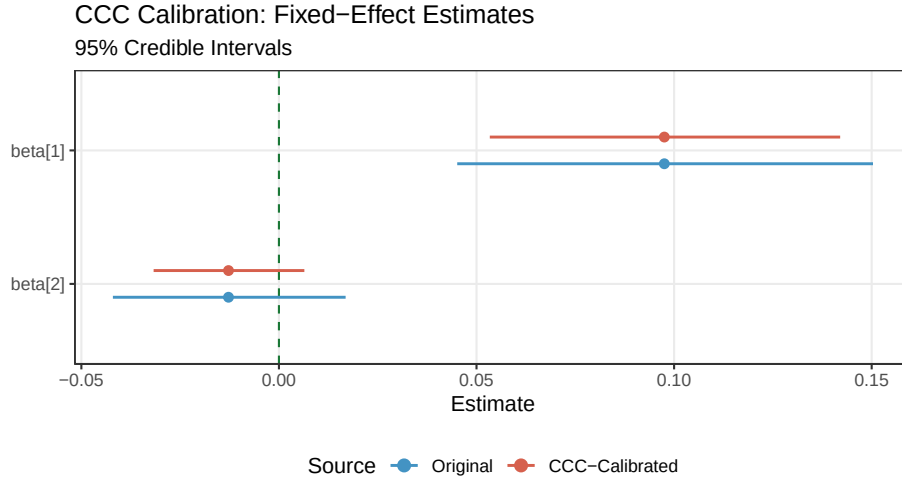


FIGURE 4. Output of `plot(cal_mod)`: original (uncalibrated posterior) and CCC-calibrated 95% intervals for the meta-regression coefficients. Centers are identical by construction (layer 1); widths change by the per-coefficient factor  $r_k$ .

(Tipton, 2015).

```
cal_raw <- ccc(fit_raw, sandwich_obj = sw_raw)
summary(cal_raw)
```

```
-- Beta Block (Cholesky CCC) -----
Frobenius ||A - I||_F: 0.999
Parameter Estimate SE    df  CI_95%_lo CI_95%_hi r_k  Direction
beta[1]   0.101   0.025 14.7  0.049     0.154   0.980 narrower
beta[2]   0.001   0.022  4.0 -0.061     0.063   2.110 wider
Beta r_k: 1 wider | 1 narrower | 0 ~same

## [... remaining blocks as before ...]
```

Two guardrails are triggered. The first is the explicit warning: the slope’s Satterthwaite  $df$  is 3.95, below the threshold of 4 that the RVE literature treats as the boundary of trustworthy  $t$ -based inference (Tipton, 2015; Tipton & Pustejovsky, 2015). Tanner-Smith et al. (2016) encountered the same situation in their own analysis of these data, where an age coefficient had  $df = 1.92$ , and recommended interpreting such coefficients with caution. The second indication is quieter but more consequential:  $r_k = 2.110$  for the slope, so the uncalibrated posterior standard deviation was less than half the robust standard error. An uncalibrated Bayesian analysis of this

specification would have reported a slope interval half as wide as can be defended. Slopes of within-study-varying moderators are the setting in which uncalibrated intervals tend to fail most severely, and also the setting in which the calibration is most valuable (Lee & Tipton, 2026a). The appropriate responses here are the ones already taken above: aggregate the moderator to the study level, as the original tutorial did, or center it within studies and estimate the within- and between-study slopes separately (Pustejovsky & Tipton, 2022), and interpret any coefficient with  $df < 4$  descriptively rather than confirmatorily.

## 7. Sensitivity to the working correlation

Users coming from `metafor`'s `vcalc()` or from `robumeta` are accustomed to assuming a within-study sampling correlation, conventionally  $\rho = .80$ , and defending that choice with a sensitivity analysis (Hedges et al., 2010; Pustejovsky & Tipton, 2022; Tanner-Smith et al., 2016). Since `sandwich()` accepts a `rho` argument with default 0.8, it is natural to ask how sensitive the results in this paper are to that choice. In fact the CCC- $t$  interval does not depend on `rho` at all. In the CE and CHE working models the cluster-level sandwich is constructed from study-mean sampling variances together with the estimated  $\tau^2$  and  $\omega^2$ , and the assumed within-study correlation cancels out of the beta block; the estimated  $\omega^2$  absorbs the within-study dependence that  $\rho$  would otherwise have to supply. The package documentation describes `rho` as its most commonly misunderstood argument for this reason. The assumed value is stored on the object for record-keeping and is passed to the frequentist cross-checks of Section 8, where it is a genuine assumption. The invariance can be confirmed directly:

```
sapply(c(0.1, 0.5, 0.9), function(r) {
  coef(sandwich(fit, rho = r)) # sandwich SE for the average effect
})
```

|  | beta[1]    | beta[1]    | beta[1]    |
|--|------------|------------|------------|
|  | 0.02224284 | 0.02224284 | 0.02224284 |

For comparison, the conventional frequentist sensitivity sweep, refitting `rma.mv()` on `vcalc()` matrices constructed at  $\rho = 0.1, \dots, 0.9$  with CR2 standard errors, moves the standard error only from 0.0226 to 0.0242 on these data (Supplement S6), reproducing the familiar finding that CR2-corrected RVE is itself nearly insensitive to  $\rho$  (Pustejovsky & Tipton, 2022; Tanner-Smith et al., 2016). The practical difference between the two routes is therefore one of bookkeeping. In the `metafor` workflow,  $\rho$

enters the working model and good practice requires the sweep to be reported; in the `bayesRVE` workflow, the reported interval involves no  $\rho$  assumption.

A genuine  $\rho$  question remains, namely what degree of within-study dependence the data themselves imply. The package addresses it with a design-effect-ratio (DER) decomposition (Lee & Tipton, 2026b): `bm_sensitivity_rho(fit)` compares the theoretical design effect implied by each candidate  $\rho$  with the empirical one, reports the implied  $\rho^*$  at which the two agree, and has a `plot()` method that displays the decomposition. For these data the empirical design effect lies at the  $\rho^* = 1$  boundary, and the function reports this with a clipping warning, indicating within-study dependence at least as strong as any admissible compound-symmetry  $\rho$  can generate. This is unsurprising for correlations computed on the same participants within longitudinal panels. These diagnostics inform interpretation but do not affect the reported intervals, and we defer the full display to the supplement (Section S5).

## 8. Interoperability with metafor and clubSandwich

The package treats the frequentist RVE software as infrastructure to connect with, in both directions.

The first bridge is a cross-check. `compare_with_cr2()` refits the model by the frequentist route, using `metafor::rma.mv()` under REML with CR2 standard errors from `clubSandwich`, and compares the results with the `bayesRVE` sandwich:

```
compare_with_cr2(fit, rho = 0.8)
```

```
i Computing bayesRVE sandwich (rho = 0.8)...
i Fitting metafor::rma.mv (REML)...
i Computing clubSandwich::vcovCR (type = "CR2")...

-- bayesRVE vs CR2 Comparison
Working correlation (rho): 0.8

-- Fixed-Effect Standard Errors
(Intercept): bayesRVE = 0.02224, CR2 = 0.02331, rel.diff = 4.6% (CR2 larger)

-- Heterogeneity (tau^2)
bayesRVE E[tau^2]: 0.008509
metafor REML tau^2: 0.008212
```

| ✓ Excellent agreement: all relative differences < 5%.

Small differences are expected, since the two routes differ in estimation (posterior means against REML) and in bookkeeping details, and the function grades the agreement. On these data all quantities agree within 5%. We suggest running this check once in every analysis and reporting its outcome; the expected magnitudes of disagreement, and the circumstances under which larger discrepancies arise, are discussed in the supplement (Section S6).

The second bridge serves users with an existing `metafor` workflow. Given a fitted `rma.mv` model, `from_metafor()` extracts the data and the variable mapping and suggests the corresponding call:

```
V <- vcalc(vi, cluster = studyid, obs = esid, rho = 0.8, data = dat)
m_che <- rma.mv(yi, V, random = ~ 1 | studyid/esid, data = dat,
              sparse = TRUE)                                # your existing model
from_metafor(m_che)
```

| ✓ Extracted 113 observations from <rma.mv> object.  
 | i Studies: 17 unique in column "studyid".  
 | i Outcome: "yi", Variance: "V".  
 | i Suggested call: `bm\_fit(yi | se(V) ~ 1, data = <your\_data>, study = ~ studyid)`

The import is deliberately conservative. When the `rma.mv` random-effects structure is richer than the CE and CHE working models, the function warns that results are not expected to match exactly, rather than attempting to replicate a structure that the package does not fit.

## 9. Reporting

A calibrated-Bayesian meta-analysis is reported like any other meta-analysis, with a small number of additional items that make the calibration reproducible and interpretable. Table 3 lists the items and where each is found in the output. For the worked example, a Results paragraph might read as follows, with bracketed numbers referring to rows of the table.

“We synthesized  $N = 113$  Fisher- $z$  correlations nested in  $J = 17$  independent samples [1]. We fit the correlated-and-hierarchical-effects (CHE) working model with the `bayesRVE` package (v0.7.0; Lee, 2026), with weakly informative default priors ( $\beta \sim N(0, 1)$ ; half- $t(3, 0, 0.5)$  on  $\tau$  and  $\omega$ ) [2–3]. MCMC used 4 chains  $\times$  1,000 sampling iterations; all  $\hat{R} \leq 1.01$ , no

TABLE 3. Reporting checklist for a bayesRVE analysis.

| Item                                                                                                       | Source in the output                                                   |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1. $N$ effect sizes, $J$ studies, effect-size metric, cluster sizes                                        | Data; <code>print(fit)</code> restates $N$ , $J$ , $K$                 |
| 2. Working model (CE or CHE) and rationale                                                                 | Analyst’s choice; see Supplement S4                                    |
| 3. Priors (defaults or custom)                                                                             | Printed in <code>print(fit)</code> ; <code>bm_prior()</code> if custom |
| 4. MCMC settings and convergence ( $\hat{R}$ , ESS, divergences)                                           | <code>bm_diagnose(fit)</code>                                          |
| 5. Heterogeneity: posterior summaries of $\tau$ (and $\omega$ )                                            | <code>summary(fit)</code>                                              |
| 6. Calibration procedure: CR2 sandwich, Satterthwaite df, CCC- $t$                                         | Header of <code>summary(cal)</code>                                    |
| 7. Per coefficient: estimate, robust SE, df, 95% CCC- $t$ CI                                               | Beta block of <code>summary(cal)</code>                                |
| 8. Calibration diagnostics: $r_k$ per coefficient; any df < 4 or envelope warnings, and the response taken | Beta block; warnings from <code>sandwich()/ccc()</code>                |
| 9. Outcome of the frequentist cross-check                                                                  | <code>compare_with_cr2(fit)</code>                                     |
| 10. Data and code availability                                                                             | Analyst’s repository; see the availability statements of this article  |

divergent transitions, and all bulk-ESS > 2,400 [4]. Posterior means were  $\tau = 0.09$  (95% CI [0.05, 0.14]) and  $\omega = 0.05$  ([0.03, 0.07]) [5]. We calibrated the fixed-effect draws with the three-layer CCC- $t$  procedure (CR2 sandwich, Satterthwaite df) [6]. The average correlation was  $z = 0.102$  ( $r \approx .10$ ), robust SE = 0.022, 95% CCC- $t$  CI [0.054, 0.149],  $df = 15.9$  [7]. Calibration contracted the interval relative to the uncalibrated posterior ( $r_k = 0.90$ ) [8]; results agreed with a frequentist CHE + CR2 analysis within 5% on all quantities [9]. Data and code to reproduce all results are available at <https://github.com/joonho112/bayesRVE-tutorial-replication> [10].”

Item 8 is the one addition relative to standard practice, and we would encourage authors to report  $r_k$  for the coefficients they interpret. A reader who sees  $r_k = 0.90$  learns that the Bayesian working model was nearly calibrated already; a reader who sees  $r_k = 2.1$  learns that the robust interval is twice the uncalibrated one and that the calibrated numbers are the ones to trust. Reporting  $r_k$  makes the calibration verifiable by the reader rather than a matter of trust in the software.

## 10. Discussion

We have described the `bayesRVE` workflow for meta-analysis with dependent effect sizes: a Bayesian CE or CHE working model is fitted with `bm_fit()`, the sampler is checked with `bm_diagnose()`, a CR2 cluster-robust target is constructed with `sandwich()`, the posterior draws are recalibrated with `ccc()`, and the resulting summary combines a Bayesian center with a sandwich-robust width and small-sample-corrected tails. On the teaching dataset of Tanner-Smith et al. (2016), the calibrated analysis reproduces the frequentist RVE conclusions where the working model is adequate, quantifies how adequate it was through the  $r_k$  ratios, identifies a specification in which uncalibrated Bayesian precision would have been overstated by a factor of two, and provides in addition the shrinkage point estimates and full posteriors for  $\tau$  and  $\omega$  that the frequentist route does not supply.

The boundaries of the package deserve restating. It fits two working models, CE and CHE, for a single level of clustering; it does not at present handle multivariate models with known within-study covariances, adjustment for publication bias (see Bartoš et al., 2026), network structures, or effect-size metrics requiring non-normal likelihoods. Calibration of the fixed effects requires  $J \geq \max(8, 2K + 2)$  studies, and coefficients with Satterthwaite  $df < 4$  should be interpreted descriptively under any framework. Within these boundaries, the combination of Bayesian estimation and robust uncertainty assessment has previously required working across two methodological literatures at once, and our aim, in the spirit of the tutorials that introduced RVE to applied audiences (Tanner-Smith & Tipton, 2014; Tanner-Smith et al., 2016), has been to make the workflow described here sufficient for a complete and reproducible analysis.

Readers who want the underlying theory can consult the companion papers in increasing order of specialization: the three-layer framework and its simulation evidence (Lee & Tipton, 2026a), the likelihood-bridge results that justify combining posterior draws with sandwich targets (Lee et al., 2026), and the design-effect-ratio diagnostics for anticipating which coefficients need correction (Lee & Tipton, 2026b). The package website (<https://joonho112.github.io/bayesRVE/>) organizes its documentation along the same applied and methodological tracks, and the Online Supplement extends this tutorial with installation notes, prior sensitivity, the CE model, the full  $\rho$  diagnostics, and a gallery of the package’s warnings.

**Software and reproducibility.** The analyses in this tutorial use the `bayesRVE` R package, version 0.7.0 (Lee, 2026), open source (MIT) at <https://github.com/joonho112/bayesRVE> with documentation at <https://joonho112.github.io/bayesRVE/>; posterior draws are obtained through Stan via the `cmdstanr` interface, and the frequentist cross-checks use `metafor` and `clubSandwich`. Documented code that reproduces every number, table, and figure in this article and its supplement (R 4.6.0, `bayesRVE` 0.7.0, seed 42) is available in the replication repository at <https://github.com/joonho112/bayesRVE-tutorial-replication>; the complete script is also listed in Supplement S2. No new data were collected and no studies were preregistered.

**Data availability.** The worked-example dataset (`dat.tannersmith2016`) and the cautionary-example dataset (`dat.assink2016`) are publicly available in the `metadat` package on CRAN (Viechtbauer et al., 2026); the quick-start dataset (`example_meta`) ships with `bayesRVE`. This tutorial reanalyzes previously published, publicly available aggregate data.

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# Online Supplement

## Bayesian meta-analysis with dependent effect sizes in R: A tutorial on the bayesRVE package

JoonHo Lee

This supplement contains material that supports, but is not required for, the workflow described in the main text: installation and troubleshooting (S1); the complete replication script (S2); prior specification and prior-sensitivity analysis (S3); the choice between the CE and CHE working models (S4); the design-effect-ratio  $\rho$  diagnostics in full (S5); extended interoperability with `metafor` and `clubSandwich` (S6); a gallery of the package’s warnings and guardrails with suggested responses (S7); and session information (S8). Section labels S1–S8 correspond to the pointers in the main text. All console displays are verbatim outputs of the replication scripts (seed 42), with elisions marked `## [...]`.

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## S1. Installation and troubleshooting

### S1.1. Installation

bayesRVE requires R ( $\geq 4.1$ ), the `cmdstanr` interface package, and a working CmdStan installation, which in turn requires a C++ toolchain. The following order works on all platforms:

```
# 1. cmdstanr (from the Stan R-universe repository)
install.packages("cmdstanr",
                  repos = c("https://stan-dev.r-universe.dev",
                           getOption("repos")))

# 2. Check the C++ toolchain, then install CmdStan itself
cmdstanr::check_cmdstan_toolchain(fix = TRUE)
cmdstanr::install_cmdstan()

# 3. bayesRVE from GitHub
pak::pak("joonho112/bayesRVE") # or remotes::install_github(...)
```

For step 2, macOS requires the Xcode Command Line Tools (`xcode-select -install` in a terminal), Windows requires the version of RTools matching the installed R, and Linux requires `g++` and `make`, which are present on most systems. `check_cmdstan_toolchain(fix = TRUE)` diagnoses and, where possible, repairs the common problems.

### S1.2. The compiled-model cache

The first call to `bm_fit()` compiles the Stan program, which takes roughly half a minute, and caches the compiled binary; later calls load it from disk in milliseconds.

The fits reported in the main text take a few seconds in total for this reason. Two helpers manage the cache:

```
bm_compile("che")      # pre-compile a model (e.g., before a workshop)
bm_clear_cache()       # wipe cached binaries (e.g., after upgrading CmdStan)
```

If fits fail with linker or ABI errors immediately after a CmdStan or compiler upgrade, clearing the cache and refitting resolves the problem.

### S1.3. If the MCMC diagnostics fail

When `bm_diagnose()` reports problems, we suggest escalating in the following order; all of the settings mentioned are arguments to `bm_fit()`. First, increase the number of iterations (`iter_warmup = 2000`, `iter_sampling = 2000`), which resolves most reports of low effective sample size. Second, request slower and more careful adaptation (`adapt_delta = 0.99`), which is the standard response to divergent transitions. Third, raise `max_treedepth` to 12 if treedepth warnings appear. Persistent divergences after these steps usually indicate that the model is a poor match for the data, as for instance when the CHE model is fitted to singleton studies (Section S7.5); in that case the working model or the priors on the variance components should be reconsidered.

## S2. Complete replication script

The following script reproduces every number, table and figure in the main text in order, in about one minute on a laptop once Stan has been compiled. It is also provided as `replication-script.R` in the replication repository (<https://github.com/joonho112/bayesRVE-tutorial-replication>), which archives the full analysis harness and its expected outputs.

```
## Reproduces all results in "Bayesian meta-analysis with dependent
## effect sizes in R: A tutorial on the bayesRVE package" (seed 42).
library(bayesRVE)      # 0.7.0
library(metadat)       # 1.6-0
library(metafor)       # 5.0.1
library(clubSandwich)  # 0.7.0

## ---- A first analysis (Section 4) -----
data(example_meta)
head(example_meta)
fit_qs <- bm_fit(yi | se(vi) ~ 1, data = example_meta,
                study = ~ studyid, model = "che", seed = 42)
```

```

sw_qs <- sandwich(fit_qs) # rho = 0.8 default
cal_qs <- ccc(fit_qs, sandwich_obj = sw_qs)
summary(cal_qs)

## ---- Application: Tanner-Smith et al. (2016) data (Section 5) -----
data(dat.tannersmith2016)
dat <- dat.tannersmith2016
head(dat, 8)
table(table(dat$studyid)) # cluster sizes

## Fit the CHE working model
fit <- bm_fit(yi | se(vi) ~ 1, data = dat,
             study = ~ studyid, model = "che", seed = 42)
print(fit)
summary(fit) # uncalibrated posterior

## Convergence diagnostics
bm_diagnose(fit)

## sandwich -> ccc -> summary
sw <- sandwich(fit, rho = 0.8)
cal <- ccc(fit, sandwich_obj = sw)
summary(cal)
coef(cal); confint(cal)

## Four analyses of the same data (Table 2)
m_ign <- rma(yi, vi, data = dat, method = "REML") # ignore dependence
dat$esid <- seq_len(nrow(dat))
V <- vcalc(vi, cluster = studyid, obs = esid, rho = 0.8, data = dat)
m_che <- rma.mv(yi, V, random = ~ 1 | studyid/esid, data = dat,
               sparse = TRUE)
clubSandwich::coef_test(m_che, vcov = "CR2", cluster = dat$studyid)
clubSandwich::conf_int(m_che, vcov = "CR2", cluster = dat$studyid)

## Meta-regression: study-mean age (Section 6)
dat$aget1_study <- ave(dat$aget1, dat$studyid)
dat$age_c <- dat$aget1_study - mean(dat$aget1_study)
fit_mod <- bm_fit(yi | se(vi) ~ age_c, data = dat,
                 study = ~ studyid, model = "che", seed = 42)
cal_mod <- ccc(fit_mod, sandwich_obj = sandwich(fit_mod, rho = 0.8))
summary(cal_mod)
plot(cal_mod) # Figure 4

## Small degrees of freedom: within-study-varying age
dat$age_raw_c <- dat$aget1 - mean(dat$aget1)

```

```

fit_raw <- bm_fit(yi | se(vi) ~ age_raw_c, data = dat,
                 study = ~ studyid, model = "che", seed = 42)
sw_raw <- sandwich(fit_raw, rho = 0.8)      # emits the df < 4 warning
cal_raw <- ccc(fit_raw, sandwich_obj = sw_raw)
summary(cal_raw)

## rho: invariance of the bayesRVE interval (Section 7) ...
sapply(c(0.1, 0.5, 0.9), function(r) coef(sandwich(fit, rho = r)))
## ... and the conventional metafor-side sweep
for (r in c(0.1, 0.3, 0.5, 0.7, 0.9)) {
  Vr <- vcalc(vi, cluster = studyid, obs = esid, rho = r, data = dat)
  mr <- rma.mv(yi, Vr, random = ~ 1 | studyid/esid, data = dat,
              sparse = TRUE)
  print(clubSandwich::conf_int(mr, vcov = "CR2", cluster = dat$studyid))
}
## implied rho and DER decomposition (S5)
sens <- bm_sensitivity_rho(fit, sandwich_obj = sw,
                          rho_grid = seq(0.1, 0.9, 0.1))
print(sens); plot(sens)

## Interoperability (Section 8)
compare_with_cr2(fit, rho = 0.8)
from_metafor(m_che)

sessionInfo()

```

## S3. Priors: specification and sensitivity

### S3.1. Specification

Priors are collected with `bm_prior()` from three distribution constructors, `normal()` and `half_t()` together with the limiting cases `half_normal()` and `half_cauchy()`, and passed to `bm_fit()` through its `prior` argument. Printing the bundle shows what will be used:

```

p <- bm_prior(beta = normal(0, 1),
              tau  = half_t(df = 3, location = 0, scale = 0.5),
              omega = half_t(df = 3, location = 0, scale = 0.5))
p

```

```

-- bayesRVE Prior Specification
beta ~ Normal(mean = 0, sd = 1)
tau  ~ Half-t(df = 3, location = 0, scale = 0.5)

```

```
| omega ~ Half-t(df = 3, location = 0, scale = 0.5)
```

These are also the defaults. Effect sizes on the standardized-mean-difference or Fisher- $z$  scale rarely exceed  $\pm 2$  in magnitude, so a  $N(0, 1)$  prior on the elements of  $\beta$  is weakly informative, and the  $\text{half-}t(3, 0, 0.5)$  priors on  $\tau$  and  $\omega$  follow the standard recommendation for hierarchical scale parameters (Gelman, 2006), being heavy-tailed enough to let the data dominate while remaining proper. For effect sizes on other scales, such as log odds ratios of large magnitude, the prior on **beta** should be widened accordingly.

### S3.2. Sensitivity analysis

A prior-sensitivity analysis refits the model under deliberately different priors and compares the calibrated results. Since each fit takes a few seconds here, the cost is negligible:

```
prior_wide  <- bm_prior(beta = normal(0, 10),
                        tau = half_t(3, 0, 0.5), omega = half_t(3, 0, 0.5))
prior_cauchy <- bm_prior(beta = normal(0, 1),
                        tau = half_cauchy(0, 1), omega = half_cauchy(0, 1))
prior_tight  <- bm_prior(beta = normal(0, 0.5),
                        tau = half_t(3, 0, 0.25), omega = half_t(3, 0, 0.25))
# refit with prior = ..., then sandwich() + ccc() as usual
```

| prior                                  | beta   | tau   | omega | CI_lo  | CI_hi  |
|----------------------------------------|--------|-------|-------|--------|--------|
| default: beta N(0,1), half-t(3,0,0.5)  | 0.1016 | 0.090 | 0.050 | 0.0544 | 0.1488 |
| wide beta: N(0,10)                     | 0.1013 | 0.090 | 0.050 | 0.0541 | 0.1485 |
| half-Cauchy(0,1) on tau, omega         | 0.1007 | 0.090 | 0.051 | 0.0535 | 0.1479 |
| tight: beta N(0,0.5), half-t(3,0,0.25) | 0.1019 | 0.089 | 0.050 | 0.0547 | 0.1491 |

Across a tenfold change in the prior scale of  $\beta$  and qualitatively different priors on the variance components, the calibrated estimate moves in the fourth decimal place and the interval endpoints by less than 0.001; with  $J = 17$  studies and 113 effect sizes, the data dominate. A table of this kind, with one line per prior, provides a compact demonstration that the conclusions do not depend on the prior. Greater prior influence should be expected when  $J$  is small (fewer than about 10 studies), when  $\tau$  is near zero, and in the singleton limit discussed in Section S7.5; these are also the situations in which the package's warnings arise.

## S4. Choosing between CE and CHE

The CHE working model, with a between-study component  $\tau$  and a within-study component  $\omega$ , is the recommended default, in line with current frequentist guidance (Pustejovsky & Tipton, 2022). The CE model estimates a single between-study component  $\tau$  and is the Bayesian analogue of the original RVE working model (Hedges et al., 2010). Fitting CE to the running example gives

```
fit_ce <- bm_fit(yi | se(vi) ~ 1, data = dat, study = ~ studyid,
               model = "ce", seed = 42)
summary(fit_ce)
```

```
-- Fixed Effects (beta) -----
Parameter   Estimate SE    CI_95%_lo CI_95%_hi Rhat  ESS
(Intercept) 0.101   0.027 0.048     0.154    1.004 601
-- Heterogeneity (tau) -----
Mean: 0.098 | Median: 0.095 | SD: 0.022
95% CI: [0.065, 0.149]
-----
```

```
summary(ccc(fit_ce, sandwich_obj = sandwich(fit_ce, rho = 0.8)))
```

```
-- Beta Block (Cholesky CCC) -----
Frobenius ||A - I||_F: 0.163
Parameter   Estimate SE    df  CI_95%_lo CI_95%_hi r_k  Direction
(Intercept) 0.101   0.023 15.9 0.052     0.149    0.837 narrower
Beta r_k: 0 wider | 1 narrower | 0 ~same
-- Theta Block (Diagonal Rescaling) -----
Statistic    Value
J (studies) 17.000
min(r)      0.006
Q1(r)       0.060
## [... remaining theta rows and verification block ...]
```

Four points of comparison with the CHE analysis in the main text are worth recording. First, the calibrated conclusions agree: CE gives 0.101 with interval [0.052, 0.149], CHE gives 0.102 with interval [0.054, 0.149], so that after calibration the choice of working model matters little for the average effect, which is the intended behavior of robust inference. Second, the accounting of heterogeneity differs: the single CE component ( $\tau = 0.098$ ) absorbs what CHE separates into a between-study part ( $\tau = 0.09$ )

and a within-study part ( $\omega = 0.05$ ), so CHE is preferable when this decomposition is itself of substantive interest. Third, sampling efficiency differed in this example: the effective sample size for the intercept was 601 under CE against 2,484 under CHE, still comfortably above the conventional threshold of 400 but a reminder to rerun `bm_diagnose()` after switching models. Fourth, the downstream objects differ in structure. Under CE the calibrated object carries a theta block, in which per-study empirical-Bayes effects are recalibrated by diagonal rescaling and the summary reports the distribution of the scaling ratios; under CHE the study effects are integrated out analytically and no theta block applies. A need for calibrated study-level effects is therefore a reason to prefer CE.

In practice we suggest CHE whenever studies contribute multiple effect sizes, and CE when calibrated study-level effects are needed, when nearly all studies contribute a single effect size (see the singleton warning in Section [S7.5](#)), or when comparability with legacy CE-based RVE analyses is wanted.

## S5. The design-effect-ratio $\rho$ diagnostics in full

The position taken in the main text is that the CCC- $t$  interval requires no assumed  $\rho$ . The package nevertheless provides a diagnostic suite for a different question: what the clustering in the data costs in information, and what degree of within-study correlation the data imply. The machinery is the design-effect-ratio (DER) decomposition developed in Lee and Tipton ([2026b](#)). For each candidate  $\rho$  the output reports three quantities:

- $\overline{\text{DEFF}}$ , the average design effect, i.e., the variance-inflation factor that clustering would impose at that  $\rho$  (for example 5.52 at  $\rho = .8$  here, so that ignoring this much dependence would understate the variance of a study mean about fivefold);
- $\text{DER}_\mu$ , the share of the design effect that falls on the average effect, as a theoretical curve in  $\rho$  and as a single empirical value estimated from the data;
- $\text{DER}_\theta$ , the share absorbed at the study level, with a conservation identity linking the pieces; the printed “gap” quantifies the departure from this identity that arises, through Jensen’s inequality, when cluster sizes are unbalanced.

```
sens <- bm_sensitivity_rho(fit, sandwich_obj = sw,
                          rho_grid = seq(0.1, 0.9, 0.1))
print(sens)
```

```

-- bayesRVE Rho-Sensitivity Analysis
Studies (J): 17 | tau2_hat: 0.007558
DER source: marginal
B_bar: 0.88 | n_j range: [2, 12]
Implied rho*: 1

-- DER vs rho
rho DEFF_bar DER_mu_th DER_mu_em DER_th_bar gap_pct mu_pct theta_pct
0.1      1.56      0.188      0.807      1.389      40.4      51.6      88.8
0.2      2.13      0.255      0.807      1.898      27.0      37.9      89.2
0.3      2.69      0.323      0.807      2.408      19.3      30.0      89.4
0.4      3.26      0.391      0.807      2.917      14.3      24.8      89.5
0.5      3.82      0.458      0.807      3.426      10.7      21.1      89.6
0.6      4.39      0.526      0.807      3.935       8.1      18.4      89.7
0.7      4.95      0.594      0.807      4.444       6.0      16.3      89.7
0.8      5.52      0.662      0.807      4.953       4.4      14.6      89.8
0.9      6.08      0.729      0.807      5.463       3.1      13.3      89.8

i rho* = 1: theoretical DER_mu matches empirical.

```

The table is read as follows. The empirical  $\text{DER}_\mu$  of 0.807 exceeds the theoretical curve at every  $\rho < 1$ , so the implied correlation  $\rho^*$  is clipped at the upper boundary: the within-study dependence in these data is at least as strong as any admissible compound-symmetry  $\rho$  can generate. (`bm_find_rho_implied(fit, sandwich_obj = sw)` returns the same  $\rho^* = 1$  with an explicit clipping warning; see Section S7.4.) This is substantively plausible, since the correlations were computed on the same participants within longitudinal panels. Two companion views complete the suite. `bm_sensitivity_rho_table(sens)` returns the conservation and budget columns as a plain data frame, and `bm_sensitivity_rho_study(fit, sandwich_obj = sw, rho = 0.8)` decomposes the design effect study by study; in these data the studies contributing  $n_j = 12$  correlations carry design effects near 9.8 and the  $n_j = 2$  studies near 1.8, which indicates which studies drive the dependence burden. Figure S1 shows the plot method: panel A displays the implied  $\rho^*$ , and panels B and C the conservation identity and the division of the design effect between the average effect and the study level.

### Sensitivity to $\rho_{\text{true}}$ : DER Decomposition

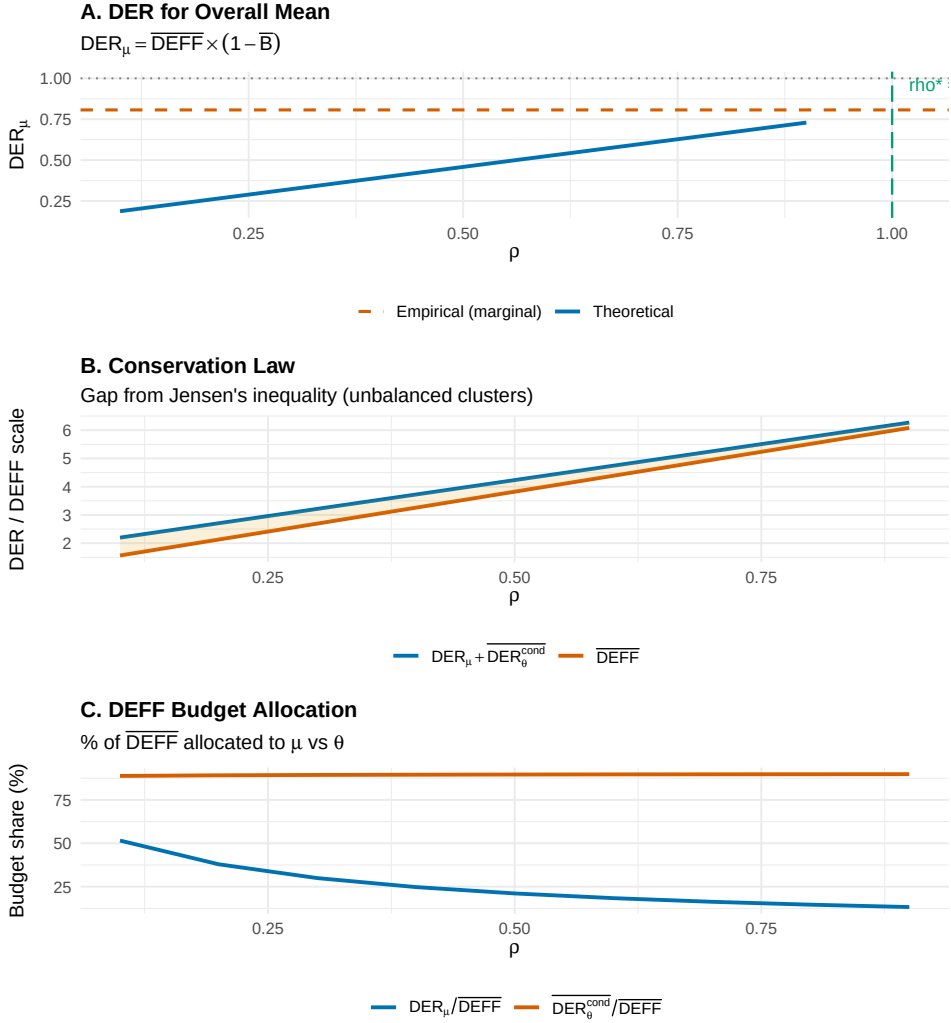


FIGURE S1. Output of `plot(sens)` for the running example: (A) theoretical and empirical design-effect ratios for the average effect, with the implied  $\rho^*$  marked; (B) the conservation identity and its Jensen gap arising from unbalanced cluster sizes; (C) the division of the total design effect between the average effect and the study level.

## S6. Extended interoperability

### S6.1. The frequentist mirror

The row labeled “CHE + CR2 RVE” in Table 2 of the main text was produced by the standard `metafor/clubSandwich` recipe, which is also the one shown on the help page for `dat.tannersmith2016`:

```

dat$esid <- seq_len(nrow(dat))
V <- vcalc(vi, cluster = studyid, obs = esid, rho = 0.8, data = dat)
m_che <- rma.mv(yi, V, random = ~ 1 | studyid/esid, data = dat,
               sparse = TRUE)
clubSandwich::coef_test(m_che, vcov = "CR2", cluster = dat$studyid)
clubSandwich::conf_int(m_che, vcov = "CR2", cluster = dat$studyid)

```

|         | Coef. Estimate | SE     | Null value | t-stat | d.f. (Satt) | p-val (Satt) | Sig. |
|---------|----------------|--------|------------|--------|-------------|--------------|------|
| intrcpt | 0.102          | 0.0241 | 0          | 4.25   | 14.9        | <0.001       | ***  |

|         | Coef. Estimate | SE     | d.f. | Lower 95% CI | Upper 95% CI |
|---------|----------------|--------|------|--------------|--------------|
| intrcpt | 0.102          | 0.0241 | 14.9 | 0.0509       | 0.154        |

## S6.2. Expected agreement

`compare_with_cr2()` automates this mirror and grades the discrepancy, as shown in Section 8. As to the magnitudes to expect, point estimates agree essentially exactly when priors are weak and  $J$  is moderate, and standard errors typically agree within 5 to 15%. The differences arise from three sources: posterior-mean as against REML estimation of the variance components; the bookkeeping of the working covariance, which is a  $\rho$ -blocked `vcalc` matrix on the `metafor` side and an  $\omega$ -absorbed diagonal structure on the `bayesRVE` side; and prior influence when  $J$  is small. Agreement within 5%, as in the present example, can be taken as mutual confirmation. Discrepancies beyond about 15% warrant investigation; in our experience they indicate that the priors are informative relative to the data, that the two model structures genuinely differ (see the `from_metafor()` warnings), or that a coefficient lies near the degrees-of-freedom guardrail.

## S6.3. The conventional rho sweep on the metafor side

For completeness, the frequentist sensitivity sweep referred to in Section 7, with `vcalc()` rebuilt at each  $\rho$  and CR2 inference throughout, gives

|  | rho | est    | SE     | lo     | hi     |
|--|-----|--------|--------|--------|--------|
|  | 0.1 | 0.1025 | 0.0226 | 0.0546 | 0.1504 |
|  | 0.3 | 0.1026 | 0.0231 | 0.0537 | 0.1516 |
|  | 0.5 | 0.1025 | 0.0235 | 0.0525 | 0.1525 |
|  | 0.7 | 0.1024 | 0.0239 | 0.0514 | 0.1533 |
|  | 0.9 | 0.1022 | 0.0242 | 0.0505 | 0.1540 |

The estimate is flat and the standard error moves by 7% over the admissible range, in line with the finding that CR2-corrected RVE conclusions are insensitive to the assumed  $\rho$  (Pustejovsky & Tipton, 2022; Tanner-Smith et al., 2016). The `bayesRVE` interval, which involves no  $\rho$  at all, sits alongside these values.

#### S6.4. Importing richer metafor models

`from_metafor()` warns when the `rma.mv` specification lies outside the CE and CHE family:

```
✓ Extracted 113 observations from <rma.mv> object.
i Studies: 17 unique in column "studyid".
i Outcome: "yi", Variance: "V".
i Suggested call: `bm_fit(yi | se(V) ~ 1, data = <your_data>, study = ~ studyid)`
```

With `struct = "CS"` random effects, for example, the function adds a note that `bayesRVE` uses a simpler random-effects structure and that results may differ. These notes should be heeded: the import bridge is intended for adopting the calibration workflow on CE- and CHE-shaped problems, not for replicating arbitrary multivariate structures.

### S7. Gallery of warnings and guardrails

Every message below is real output, reproduced by the S7 section of the supplement script. The package computes only what the data can support and states explicitly when it declines to do more; this section describes each message and the response we suggest.

#### S7.1. Operating envelope: too few studies

With  $J < J_{\min} = \max(8, 2K + 2)$ , the beta block falls back to the raw posterior. Here the CE model is fitted to the first 6 studies of the running example:

```
i J = 6 < J_min = 8 (operating envelope).
i Beta-block CCC will use raw posterior (D7: beta-only fallback).
i Theta-block CCC proceeds normally.

## [... fit and further fallback notes ...]

-- Beta Block (Cholesky CCC) -----
! Beta-block CCC was skipped: sandwich_fallback: J_too_small
Showing original (uncalibrated) beta estimates.
Parameter   Estimate SE    df CI_95%_lo CI_95%_hi r_k Direction
```

```
| (Intercept) 0.073    0.072 NA -0.072    0.212    NA skipped
```

A cluster-robust covariance estimated from six clusters is too noisy to serve as a calibration target, so the package displays the uncalibrated posterior and says so; the df and  $r_k$  entries are NA and the summary header changes to “Inference: CCC (none)”. These intervals should not be reported as robust. The available options are to accept model-dependent Bayesian inference and state as much, to reduce  $K$ , or to obtain more studies. The verification line reads “3/6 checks passed (3 skipped: beta fallback)”, and in this situation, unlike the  $K = 1$  case discussed in Section 4, the fallback label is literal.

## S7.2. Small Satterthwaite degrees of freedom

```
| Warning: ! Satterthwaite df = 3.95 < 4 for coefficient 2; inference may be unreliable
| (Tipton, 2015).
```

The effective number of clusters supporting this coefficient is small enough that the  $t$  approximation is at the boundary of its validity (Tipton, 2015). The interval should be treated as descriptive, and the moderator simplified, aggregated to the study level, or recoded with fewer categories, as discussed in Section 6.

## S7.3. CR2 adjustment near singularity

From the cautionary example in Section S7.6:

```
| Warning: ! Study 16: 1 eigenvalue(s) of (I - H_tilde) below 1e-10; truncated.
|i CR2 adjustment may be unreliable for this study.
```

One study exerts so much leverage, typically by dominating a moderator category, that the CR2 bias-correction matrix is close to not existing. This is a problem of the data design rather than a numerical nuisance; see Section S7.6.

## S7.4. Implied correlation outside [0, 1]

```
| Warning: ! rho_implied (1.015) falls outside [0, 1]; clipping to boundary.
|i This indicates the empirical DER is outside the theoretical range achievable for
| any rho in [0, 1].
```

This message is diagnostic rather than an error: the dependence in the data exceeds what compound symmetry can represent (Section S5). No action is required for inference, since the CCC- $t$  interval does not use  $\rho$ ; substantively, the message is to be expected whenever effect sizes derive from the same participants.

## S7.5. Singleton studies

Aggregating the running example to one averaged effect size per study and fitting CHE gives:

```
Warning: ! All 17 studies are singletons (k_j = 1).
i `tau2` and `omega2` are not separately identified.
i Posterior separation relies entirely on the prior.
i CR2 benchmark may diverge from a multi-level frequentist analysis.
```

With one effect size per study the data contain no information for separating heterogeneity into between-study and within-study parts, and the posteriors of  $\tau$  and  $\omega$  simply reproduce the prior. The CE model, with its single  $\tau$ , should be used for singleton or near-singleton data; a milder version of this warning appears when more than half of the studies are singletons.

## S7.6. A cautionary example: a degenerate target

The guardrails matter most when several are triggered at once. The `dat.assink2016` dataset (Viechtbauer et al., 2026) comprises 100 standardized mean differences from 17 studies, each contrasting recidivism between delinquent juveniles with and without a mental health disorder. It contains a three-level moderator `delttype`, the type of delinquent behavior in which juveniles could have recidivated, whose categories are highly unbalanced, with 78, 13 and 9 effect sizes concentrated in few studies. Fitting CHE with this moderator and calibrating produces

```
Warning: ! Study 16: 1 eigenvalue(s) of (I - H_tilde) below 1e-10; truncated.
i CR2 adjustment may be unreliable for this study.
Warning: ! Satterthwaite df = 1.99 < 4 for coefficient 2; inference may be unreliable
(Tipton, 2015).
Warning: ! Satterthwaite df = 1.74 < 4 for coefficient 3; inference may be unreliable
(Tipton, 2015).

-- Beta Block (Cholesky CCC) -----
Frobenius ||A - I||_F: 34110.739
Parameter Estimate SE      df  CI_95%_lo  CI_95%_hi r_k      Direction
beta[1]   -0.177   4351.607 11.9 -9486.743  9486.389 19694.628 wider
beta[2]    0.660   4351.615  2.0 -18826.997 18828.317 22343.273 wider
beta[3]    0.387   4351.597  1.7 -21694.008 21694.782 21529.245 wider
```

Here the CR2 target itself is degenerate: the near-singular adjustment for the dominating study inflates the robust covariance beyond meaning, and the calibration

faithfully matches the draws to that meaningless target. This is also why the internal verification still passes, since the transformation did exactly what was asked of it; the implausible magnitudes are the signal that something is wrong. A table such as this should not be reported. The small degrees of freedom (2.0 and 1.7) and the eigenvalue warning had already indicated that the distribution of this moderator across studies cannot support cluster-robust inference, and the frequentist route fails on the same specification for the same reason (compare the df guidance in Tanner-Smith et al., 2016). The sparse categories should be merged or recoded, or the moderator examined descriptively. The example also clarifies the status of  $r_k$  as a diagnostic ratio: values of this magnitude indicate a degenerate target, not a large but meaningful correction.

### S7.7. Benign Stan messages

```
Chain 4 Informational Message: The current Metropolis proposal is about to be
rejected because of the following issue:
Chain 4 Exception: normal_lpdf: Location parameter[1] is inf, but must be finite!
Chain 4 If this warning occurs sporadically, such as for highly constrained
variable types like covariance matrices, then the sampler is fine, ...
```

Occasional rejected proposals during early warmup are a normal part of adaptation, as the message itself states. No action is needed when they are sporadic and `bm_diagnose()` is clean afterwards; investigation is warranted only if the message floods the log or the diagnostics fail.

## S8. Session information

All results in the article and in this supplement were produced with R 4.6.0 (R Core Team, 2026) and the package versions below, with seed 42 throughout; the complete `sessionInfo()` output is included in the replication repository.

```
R: R version 4.6.0 (2026-04-24)
bayesRVE : 0.7.0
metadat : 1.6.0
metafor : 5.0.1
clubSandwich : 0.7.0
cmdstanr : 0.8.0
CmdStan: 2.38.0
```

Rerunning on a different platform or BLAS may perturb MCMC-derived quantities in the third decimal place; the qualitative statements, and every quantity not derived from MCMC (the frequentist rows of Table 2 and the `vcalc` sweep), reproduce exactly.